

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter 2023 Course: B. Tech. Branch : Mechanical Engineering/Automobile Engineering/Production Engineering Semester : VI Subject Code & Name: Energy Conservation and Management (BTMOE605C_Y22) Max Marks: 60 Date: 25/01/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All Questions are Compulsory 2. Draw neat diagram wherever necessary. 3. Figures to right indicates full marks 4. Assume suitable data wherever necessary and mention it clearly			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Define 1.Commercial and Non Commercial energy 2. Renewable and Non-renewable energy	R/CO 1	06
B)	Explain Energy Conservation and its importance	U/CO 1	06
C)	What is energy management?	U/CO 1	06
Q.2	Solve Any Two of the following.		
A)	Explain in brief the difference between preliminary and detailed energy audits?	U/CO 2	06
B)	Explain use of following devices i. Lux meter ii. Tachometer iii. Manometer	R/CO 2	06
C)	What are the benefits of benchmarking energy consumption?	A/CO 2	06
Q. 3	Solve Any Two of the following.		
A)	Explain simple payback period with advantages and disadvantages. Calculate the payback period for following example- A chemical plant is planning to install continuous deodorizer costing Rs. 60 Lakh for purchasing and installation. It is going to cost Rs. 1.5 Lakh per year for its operation and maintenance. This deodorizer is expected to save Rs. 20 Lakh per year through saving the steam consumption as compared to batch deodorizer.	A/CO 3	06
B)	Explain energy conservation techniques in Steam and Condensate systems.	R/CO 3	06
C)	Explain i) Payback period ii) Net Present Value	U/CO 3	06
Q.4	Solve Any Two of the following.		
A)	What is meant by combined cycle cogeneration?	U/CO 4	06
B)	Explain the term topping cycles with examples.	R/CO 4	06
C)	List and explain kinds of thermal insulation.	R/CO 4	06
Q. 5	Solve Any Two of the following.		
A)	Explain the concept of power factor in electrical system, with its mathematical formula. State the benefits of power factor improvement.	U/CO 5	06

B)	Explain energy conservation in illumination systems.	U/CO 5	06
C)	Explain energy conservation in industries.	U/CO 5	06
	*** End ***		